Guess Papers by M.N.A. GHUMMAN

GUESS PAPER 2021 CLASS: **F.Sc** SUBJECT: **Biology**

Guess Papers by:

MNA GHUMMAN

Smart Syllabus

پنجاب کے سب سے کامیاب اور مقبول ترین گیس پیپرز

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13 سالہ تدریسی تجربے کا نچوڑ

CLASS: F.Sc

SUBJECT: Biology

These Guess Papers are best for:

LAHORE, GUJRANWALA, RAWALPINDI, SARGODHA, FAISALABAD, SAHIWAL, MULTAN, DG. KHAN, BAHAWALPUR, AZAD KASHMIR BOARDS

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F.SC PART 2 BIOLOGY PAPER SCHEME

	Q#	U#15	U#16	U#17	U#18	U#19	U#20	U#21	U#22	U#23	U#24	U#25	U#26	U#27
MCQ	Q#1	2	2	1	2	1	1	2	1	1	1	1	1	1
SQ	Q#2	3	3		2								2	2
	Q#3			3					3	3		3		
	Q#4					2	3	2			2			
LQ		Q:5a	Q:6a	Q:7a	Q:8a	Q:9a	Q:6b		Q:8b		Q:9b	Q:5b		Q:7b

Most Important MCQ's

1. Flame cells are part of excretory system of: Planaria
2. The incidence of calcium oxalate type stones are: 70%
3. Human abdominal cavity is lined by a thin epithelium called: Peritoneum
4. Which part of the plant body serves as excretophores? Leaves
5. The excretory product that requires maximum water for its removal is: ammonia
6. The more concentrated external environment is: Hypertonic
7. 1g of ammonia nitrogen requires how much water for excretion: 500ml
8. Nitrogen waste us very toxic and dissolves quickly in body fluid is: ammonia
9. The homeostatic thermostat is present in: Hypothalamus
10. The membrane that bounds vacuole: tonoplast
11. Turgor pressure is generated by high osmotic pressure of the cell: vacuole
12. Angular thickenings in their primary walls are present to:Collenchyma
13. The movement shown by sperm of Liverworts towards archegonia is called: chemotactic
14. Mature bones cells called: Osteocytes
15. The number of Lumbar vertebrae are: 05
16. Which is not a joint disease: arthritis
17. Nociceptors produce sensation of: pain
18. During non- conducting state, the neuron membrane is permeable to efflux of: K^+
19. The simplest form of learning behavior is:Habituation
20. Reproduction is very important to the survival of: species
21. Developing seeds are a rich source of: Auxin+ Gibberellins+ Cytokinesis
22. Parthenocarpy is sometimes artificially induced in tomamto, peppers etc, by adding: Auxin
23. Fruit ripening is due to the production of: Ethene
24. Vehicle for transport of male gamete in land plants: pollen tube
25. Germinating pollen grain is a rich source of: Auxin
26. Fruit ripening is often accompanied by burst of respiratory activity called: Climacteric
27. Evolution of pollen tube is parallel to the evolution of: Seed
28. Which is a long day plant? Henbane

29. Diploid parthenogensis may occur in: Aphids
30. In honey bee, males are haploid and produce sperms by: Mitosis
31. B/w the semininferous tubules are interstitial cells, which secrets: Testosterone
32. Decrease of FSH and increase of estrogen causes the pituitary gland to secrete: LH
33. Which harmone induces labour pain? Oxytocin
34. Average loss of blood during birth in human female: $350cm^3$
35. In human beings, most of the major organs of embryo are formed within the: ten weeks
36. Syphilis is caused by aspirochete named as: treponema pallidum
37. The negative physiological changes in our body are: aging
38. For maximum growth, optimum temperature is: 25-30^oC
39. The shell, over chick egg is secreted as it passes through: Ovary
40. The cavity formed b/w somatic and splanchnic mesoderm is: Coelem
41. Grey equatorial cytoplasm gives rise to: notochord and neural tube
42. Gray vegetal cytoplasm gives rise to: Gut
43. Clear cytoplasm produces: Larval epidermis
44. Branch of biology which deals with the study of abnormal development & their cause: Teratology
45. The particular array of chromosomes that an individual possess: Karyotype
46. Transfer of genetic material from one cell to other that can alter the genetic make up of recipient cell is called: Transformation
47. Human cells contain types of tRNA molecules:45
48. Which of the following is a non sense codon: UGA
49. The period of life cycle of cell b/w two consecutive divisions: inter phase
50. Post mitotic cells can exist the cell cycle during:G_1- phase
51. Chromosomes doubling occurs in: S-Phase
52. Phragmoplast is formed by vesicles originated from: Golgi complex
53. Chromatin network is visible during: metaphase
54. During cell division, the nuclear division is: Karyokinesis
55. Cancer is caused mainly by mutation in: somatic cells
56. Full cell cycle in yeast cells has length of: 90 minutes
57. The microtubules are composed of a protein tubulin and traces of: RNA
58. The most critical phase of mitosis, which ensures equal distribution of chromatids in the daughter cells is: Anaphase
59. Crossing over occurs during: Pachytene
60. In which stage of Meiosis, the paired chromosomes repel each other and begin to separate: Diplotene
61. Different alleles of a gene that are both expressed in a heterozygous conditions: Co dominant
62. The position of gene one chromosomes: Locus
63. Expression of a trait: phenotype
64. ABO blood group system was discovered by:Karl Landsteiner

65. A full set of genes of an individual: Genome
66. Cystic fibrosus patients lack a gene that codes for trans-membrane carrier of: chloride ion
67. The enzyme which are used to cut out the gene of interest are: Restriction endonuclease
68. The polymerase are enzyme present in: Bacteria
69. The genome fragmant can be separated according to their lengths during: Gel electrophoresis
70. Archaeobacteria can tolerate temperature upto:120°C
71. Lyell published the principle of: Geology
72. Armadillos, the armored mammals live only in: America
73. A group of individuals belong to a particular species and sharing a common geographic are: Population
74. The change in frequency of alleles at locus that occur by chance: Genetic drift
75. Endangered species of plants have been recorded to more than: 500
76. Who proposed the term, “Niche” in ecology: Josph Grinnel
77. All the food chains and food webs begins with: produces
78. Herbaceous stage in xerosere is the: fourth stage
79. Succession begins by a few hardy invaders: Pioneers
80. The remoras benefit from the relationship with shark: Commensalism
81. Disease in living organism is caused by parasites are: Infestation
82. When bacteria in soil oxidize ammonia or ammonium ion: ammonification
83. The total energy of the sun , trapped by the producer in an ecosystem is about: 01%
84. All the population within an ecosystem : community
85. The biomes which has been increased in area by human activities: desert
86. The average rainfall in temperate deciduous forests is b/w: 750-1500mm
87. Coniferous forest located at high latitude: Boreal
88. Northern coniferous forest: taiga
89. Coniferous forest located at high altitude: alpine
90. In temperate grassland the rate of primary production:700-1500 g/m^2
91. Desert ecosystem of Bhakkar and Mianwali: Thal
92. Felis bengalensis is the zoological name of:Leopard cat
93. Study of human populations and things that affect them: Demography
94. At the time of independence, the population of Pakistan was about: 32.5 million
95. Which of the following acts as environment buffer is a: forest
96. Establishment of new forests, where no forests existed before is: afforestation
97. The decline in thinkness of ozone layer is caused by increasing level of: chloroflurocarbon
98. Ozone in the upper layer of atmosphere that filters: UV raduation
99. Total area of the world under cultivation: 11%
100. The befouling of the air by anything that may be harmful to living organism: air pollution

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Most Important Short Questions

Unit 15

1. Define homeostasis.
2. What are mesophytes. Give examples.
3. Give four adaptations of Xerophytes.
4. Differentiate b/w osmoconformers and osmoregulation.
5. What is Anhydrobiosis?
6. Sketch metabolic pathway of urea cycle.
7. Give structure of protonephridium.
8. Why leaves are said to be excretophores.
9. Differentiate ureotely from uricotely.

Unit 16

1. Differentiate b/w Collenchyma and Sclerenchyma cells.
2. Differentiate b/w fibers & sclereides.
3. Differentiate b/w Epinasty and Hyponasty.
4. Differentiate b/w Hyaline and elastic cartilage.
5. What is Ecdysis or moulting? Give examples.
6. What is rickets, give its causes.

Unit 17

1. Give commercial application of Ethene.
2. Differentiate b/w axons and dendrites.
3. What are effecters? Give example.
4. What is resting membrane potential? Give its value.
5. Differentiate b/w sympathetic and parasympathetic system.

Unit 18

1. Define parthenocarpy with examples.
2. What is seed dormancy? Give its importance.
3. Define vernalization. Discuss its importance in plants.
4. What are apomixes?
5. What is tissue culture? Explain.
6. How identical twins are produced?
7. Explain Oestrous cycle?
8. What is meant by menopause?
9. What are test tube babies?

Unit 19

1. Differentiate b/w growth and development.
2. Write different types of plant meristems?
3. Define apical dominance.
4. What is Blastoderm? Name its layers.
5. Name two layers of lateral plate of mesoderms.
6. Differentiate b/w Neural and Neurocel.

Unit 20

1. What is Karyotype? Give its importance.
2. What is nucleosome? What charge is present on histone?
3. Differentiate b/w Heterochromatin and Euchromatin.
4. What is phsphodiester bond?
5. Differentiate b/w transcription and replication.
6. Define central dogma.
7. Compare template strand with coding strand.
8. Define translation.

Unit 21

1. Describe cell cycle. Draw and label it.
2. What is mitotic apparatus? What is its role during cell division?
3. What are cytokinesis and karyokinesis?
4. How cytokinesis takes place in plants?
5. Define metastasis? How it occurs?
6. What is malignant tumor?
7. Describe non-disjunction.
8. What is Turner's syndrome?

9. Differentiate b/w necrosis and apoptosis.

Unit 22

1. Differentiate b/w phenotype and genotype.
2. What is test cross? Give its application.
3. Differentiate b/w complete and over dominance.
4. What are polygenes? How polygenic inheritance takes place?
5. Differentiate b/w sex chromosomes and autosomes.
6. What is a sex-limited trait?
7. What is MODY?

Unit 23

1. What are restriction enzymes?
2. What is palindromic sequence and sticky ends.
3. What is probe? Give its uses.
4. What are totipotents?

Unit 24

1. Define theory of special creation.
2. What are hydrothermal vents?
3. Explain endosymbiont hypothesis.
4. Differentiate b/w homologous and analogous organs.
5. What are vestigial organs? Name some of them.
6. Define genetic drifts and give its effects.
7. Give two ways to protest endangered species.
8. Give two ways to protect endangered species.

Unit 25

1. Differentiate b/w population and community.
2. Define biosphere and niche.
3. Differentiate b/w synecology and auteocology.
4. Define food chain and food web.
5. What are decomposers?
6. Differentiate b/w hydrosere and xerosere.
7. Define succession.
8. What are lichens? Give examples.
9. What is commensalism with example?
10. Give grazing.

Unit 26

1. Differentiate b/w limnetic and profundal zone?
2. Differentiate b/w aline and boreal forests.
3. Differentiate b/w praries and savanna.
4. What is desert ecosystem located in Pakistan?

Unit 27

1. Differentiate b/w renewable and non renewable resources.
2. Define demography.
3. What is green house effect?
4. How algal bloom pollution may occur?
5. Write a note on acid rain.
6. Differentiate b/w deforestation and afforestation.

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Most Important Long Questions

1. **(a).** Write a note on osmoregulation in marine animals. **Unit 15**
   (b). Explain excretion in plants. **Unit 15**
2. **(a).** Explain urinary system of human. Describe the structure of nephron. **Unit 15**
   (b). Discuss kidney problems & their cures. **Unit 15**
3. **(a).** Describe types of paratonic movements in plants. **Unit 16**
   (b). What are joints? Explain their types. **Unit 16**
4. **(a).** Give structure of skeletal muscles. **Unit 16**
   (b). Define & explain nerve impulse. Describe the initiation of nerve impulse. **Unit 17**
5. **(a).** Define central nervous system. Explain structure & functions of different parts of human brain. **Unit 17**
   (b). Discuss role of harmones secreted by anterior lobe of pituitary gland. **Unit 17**
6. **(a).** Define parthenogenesis. Explain its kinds. **Unit 18**
   (b). Explain human female reproductive system. **Unit 18**
7. **(a).** Give an account of sexually transmitted disease in man. Also write the causes & effects of sexually transmitted disease. **Unit 18**
   Discuss external factors affecting growth in plants. **Unit 19**
8. **(a).** Write a note on aging. Describe its causes and symptoms. **Unit 19**
   (b). Write a note on abnormal development. **Unit 19**
9. **(a).** Discuss Watson and Crick's model of DNA. **Unit 20**
   (b). Explain Meselon-Stahl experiment. **Unit 20**
10. **(a).** Write a note genetic code. Describe its characteristics. **Unit 20**
    (b). Define and explain test cross. **Unit 22**
11. **(a).** Explain different patterns of sex determination. **Unit 22**
    (b). Discuss genetics of colour blindness. **Unit 22**
12. **(a).** Explain Darwin's theory of natural selection. **Unit 24**
    (b). Describe the evidence of evolution from comparative anatomy. **Unit 24**
13. **(a).** Describe various factors affecting gene frequency. **Unit 24**
    (b). What are endangered species? What measures could be adopted for their environment protection? **Unit 24**
14. **(a).** Define ecosystem. Describe various components of an ecosystem. **Unit 25**
    (b). Describe different stages of succession in Xerosere. **Unit 25**
15. **(a).** Write a note on grazing. **Unit 25**
    (b). Define biochemical cycles. Explain nitrogen cycle with the help of diagram. **Unit 25**
16. **(a).** Briefly discuss sources of energy. **Unit 27**
    (b). Write a note on green house effect & acid rain. **Unit 27**
17. **(a).** Differentiate b/w Insecticide & pesticide. **Unit 27**